

Διόφαντος ὁ Ἀλεξανδρεύς

Arithmetic, Geometry, and the Work of Diophantus of Alexandria

eul_aid: rqm

Period: [Ῥωμαϊκή](#) [Roman](#)
Dialect: [Κοινή τεχνική](#) [Technical Koine](#)
Domain: [Ἱστορία](#) [History](#)
Format: [Πεζός](#) [Prose](#)

Diophantus of Alexandria was a Greek mathematician active in Alexandria during the 3rd century CE. The dating of his life relies on a dedication in his *Arithmetica* to a "Dionysius," often identified with St. Dionysius of Alexandria, placing Diophantus around 250 CE. He worked within the Hellenistic mathematical tradition in a major center of Roman-era Greek learning.

His principal work is the *Arithmetica*, a treatise on algebra and number theory originally in thirteen books. Six books survive in Greek, and four further books exist in an Arabic translation, their relation to the original being debated. The work is a collection of problems leading to determinate and indeterminate equations, now called Diophantine equations. He also authored *On Polygonal Numbers*, of which a fragment survives.

Diophantus is a seminal figure in the history of algebra. His *Arithmetica* introduced a syncopated algebraic notation and a systematic approach to solving equations, marking a departure from classical geometric algebra. His work on indeterminate equations became central to number theory, influenced Byzantine and Islamic mathematics, and was crucial to the development of algebra in Renaissance Europe.

Works (5)

- [Ἀπόσπασμα · Astronomical Calculation Methods](#) · 13 passages
- [Ἀποσπάσματα περὶ Κύκλων Γεωμετρικά · Geometric Fragments on Circles](#) · 17 passages
- [Ἀπόσπασμα · On Extracting Square Roots](#) · 1 passages
- [Περὶ Πολυγωνικῶν Ἀριθμῶν · On Polygonal Numbers](#) · 16 passages
- [Τὰ Ἑξ Βιβλία περὶ Ἀριθμητικῆς · Six Books-Arithmetic](#) · 224 passages